House Price Prediction – Project Summary

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Project Title:-House Price Prediction Using Machine Learning

- Project Objective

This project aimed to create a machine learning model that can predict house prices based on different property features. These features include area, number of bedrooms, bathrooms, parking spaces, air conditioning, and furnishing status. The project included data exploration, preprocessing, model training, evaluation, and visualization.

- Dataset

The Housing dataset was loaded using the Pandas library. During the exploration phase, we looked for missing values and duplicate records. We found no missing values or duplicate rows. Categorical columns were converted to numerical format using one-hot encoding to prepare the data for machine learning models.

- Models Used

• Linear Regression
• Random Forest Regressor

The dataset was split into 80% training data and 20% testing data using the train-test split method.

- Model Performance

The Linear Regression model performed better than the Random Forest model.

Linear Regression Results

• Mean Absolute Error (MAE): 970,043.40
• Root Mean Squared Error (RMSE): 1,324,506.96
• R^2 Score: 0.6529

Random Forest Results

• Mean Absolute Error (MAE): 1,021,546.04
• Root Mean Squared Error (RMSE): 1,400,565.97
• R^2 Score: 0.6119

- Data Visualization

We created three visualizations:

1. Distribution of House Prices (Histogram)
2. Correlation Heatmap
3. Actual vs Predicted House Prices Scatter Plot

The heatmap showed that features like area, bathrooms, air conditioning, stories, and parking had the strongest positive relationship with house prices.

- Conclusion

This project effectively showed how machine learning can help estimate house prices based on property features. Among the two models, Linear Regression provided better prediction accuracy for this dataset. This work emphasizes the importance of data preprocessing, feature analysis, and model evaluation in building reliable predictive models.